José L. Pabón, PhD

PhD mathematician, published researcher, data scientist, certified enterprise carrier grade IT DevOps.
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PERSONAL STATEMENT

My academic preparation in **computational mathematics** coupled with my training and vast experience in enterprise, carrier grade **IT DevOps** allow me to operate nimbly to navigate the challenges of unearthing the unknown insights in **Data Science** and Technology. With my **PhD** from NJIT Rutgers, via A.B. from Princeton University, I look to apply the rigorous, data driven methodologies acquired at these R1 research institutions to new roles and opportunities. My **technology industry background** has provided me with invaluable **quantitative and qualitative skills**. The ability to solve complex problems with advanced models is the table stakes of our labor market; leveraging these solutions, via **data visualization and storytelling** that stakeholders can understand, is the key to **our shared success**.

EDUCATION

Ph.D. in Mathematical Sciences, **New Jersey Institute of Technology.** Graduation date: August 2025
A.B. Degree in Mathematics, **Princeton University.** Information Tech Policy, PLAS Minor, Graduation date: June 2019

SKILLS

Languages: *Computer Science:* Native in Matlab, LaTeX, Fluent in HTML, Markdown and Hugo I/O, built on Go. Experienced in Java, Python, and R- Studio. *Human*: Native language proficiency in English, Spanish, Fluent in French.
Data Science skills highlights: Optimization, High Dimensional Dataset Wrangling and Analysis, Machine Learning, Physics informed forward neural networks, Binary classification analysis, Discrete and Continuous Dynamical System modeling experience, apps to evolution of dynamical systems, k-means clustering analysis, Python scikit. Expert MS Excel, Word, etc.
Industry and Technology Certifications:
BroadWorks Certified Operational Engineer, Voice over I.P. and Cloud-based unified communications and collaboration focused. Computing Technology Industry Association (CompTIA) Network Plus Administrator. AT&T Six Sigma program Orange Belt, Green Belt.
AT&T Enterprise Carrier Grade Technology skills and management experience:

- Carrier grade enterprise systems and equipment administration, including:
- Oracle Acme-Packet Session Border Controllers model 4600, 3850, 3250.
- Nortel Passport Switch model 8006 and 8010.
- VMWare ESXi virtual platform on IBM hardware; Packet tracing (Wireshark) of TCP/IP and VoIP datagrams.
- Nagios Icinga SNMP trap monitoring, as well as fault monitoring on IPv4, IPv6 and Cisco nets.
- Cisco Ironport model c170 and c370, mail SORBS blacklist and whitelist reputation management.
- Windows active directory, group and org policy, DNS, DHCP server configs, via PowerShell scripting

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EXPERIENCE

- ***Doctoral Researcher*, W. River Watershed Council (WRWC), RIOS Institute,** Fall 2025 through present
  - Coordinate, investigate and develop novel results for partner environmental and community nonprofit 501(c)(3) charity.
  - VECINITA bespoke large language model project. Specs: Groq (Llama 3.1) LLM, HuggingFace all-MiniLM Embed, Supabase PostgreSQL DB, FastAPI Python 3.10 Backend, Static HTML/JS Frontend, LangChain BeautifulSoup + Playwright Scraping, Docker on Google Cloud VM Deployment. Also GIS enabled flooding heat map shiny apps deploy.

- ***Research Assistant & Laboratory Instructor*, New Jersey Institute of Technology,** Fall 2019 to Summer 2025
  - Develop theoretical fluids dynamical models conceptualizing physical concepts and implementing efficient, object oriented, modular scalable algorithm based code for the numerical analysis of the models.
  - Managed software including Matlab, Python, LaTeX, High Performance Computing Optimization.
  - Designed the laboratory, seminar and recitation content independently for undergraduate and graduate courses. Graded exams, problem sets and projects and assigned student performance via Canvas platform.

- ***Visiting Researcher*, Brown University, Institute for Computational and Experimental Research in Mathematics (ICERM),** Concurrent appointment May to July 2023
  - Collaborated with investigative faculty, used Data Science techniques to document inequities in communities.

- ***Visiting Researcher*, University of Chicago, Institute for Mathematical and Statistical Innovation.** March to June 2024

- Project management, realization of VECINA – Data Science (GIS) web tool for the Woonasquatucket River Watershed Council (WRWC), including Git repo, production Application Programming Interface (API) docs.

- ***Visiting Researcher*, University of Edinburgh, Edinburg Futures Institute: Math for Humanity Program,** Concurrent appointment June to August 2024
  - Led two parallel workgroups in research, including Python sci kit machine learning models, to validate and operationalize our novel Providence-Edinburgh Genocide Resilience Guidelines. See publications for more details.

- ***Founder and Technology Officer*, Reclaim Energies,** Spring 2018 through Summer 2019
  - Originated green hydroelectricity company, backed by Princeton U. eLab Keller Center. Composed tech docs.

- ***DevOps Datacenter Administrator* AT&T,** Spring 2000 through Summer 2018
  - Oversight of 500+ server, network connectivity, HVAC environment and electrical connections in 10,000 sq. ft. datacenters as exempt Dev Ops management employee with direct reports both contractors and internal.
  - Management of N+2 data redundancy , HVAC and power infrastructure, regulations compliant ISACA,FISMA HIPAA.
  - Authored method of operations procedure documentation; led various interdepartmental projects cradle to grave.
  - Real time incremental and full backup solution EMC Dell Avamar implementation via high speed Storage Area Networks; audited backups and their policies to ensure data integrity and datastore optimization.
  - Tier 4 highest level escalation contact for troubleshooting carrier grade telecommunications equipment.

SELECTED PROJECTS, PUBLICATIONS, & RELATED DATA SCIENCE SKILLS

(Published) Buckmire R*, **Pabón JL**, et al. *(2025). Quantifying and Documenting Gender-Based Inequalities in the Mathematical Sciences in the US in Advances in Data Science: Women in Data Science and Mathematics (WiSDM) 2023 (pp. 335-352). Cham: Springer Nature Switzerland.* In Advances in Data Sciences, Springer Ed. Lee, H and Garcia-Cardona. URL , Skills used: Data modeling, validation inc. chi squared, Kendall Tau testing, visualization.
(In press) Piercey V*, Greene A*, **Pabón JL*,** et al., Edinburgh-Brown Framework: *Data Informed Machine Learning Genocide Detection.* In La Matematica, Springer Ed. Bañuelos, S, Danielli, D, Leonard, K, Manes, M and Radunskaya, A. Skills used: Data modeling, Python scikit machine learning data validation, k-means clustering.

(In press) Buckmire R*, Hibdon JE, **Pabón JL**, et al., *The Mathematics of Mathematics: Using Mathematics and Data Science to Analyze the Mathematical Sciences Community and Enhance Social Justice*. In La Matematica, Springer Ed. Bañuelos, S, Danielli, D, Leonard, K, Manes, M and Radunskaya, A. Preprint link Skills: Data modeling, testing.

(Preprint) **Pabón JL*,** Oza, A, & Siegel, *M. Reduced order, computationally efficient models for hydrodynamically interactive collective motion.* Dissertation. Skills: High dimensional object oriented code, data modeling, analysis

(Published in excerpt) **Pabón JL*,** Racz, M, & Coda Marques, F. *A Game of Prones: How prone are cryptocurrency users to fraud? An expanded game theory model of the cryptocurrency mining market.* In The Tortoise, Princeton, Spring 2019. Fancy,B and Irwin Wilkins, A. URL link. Skills: Cryptography, cryptocurrency, data analysis and modeling.
***** - *Denotes* ***main author*** *of correspondence.*

SEMINARS & PRESENTATIONS:

- 2023 Society for Industrial and Applied Mathematics (SIAM) Conference on Computational Science and Engineering (CSE23), RAI Amsterdam; MS23 Symposium on Numerical Simulation of Bio-Inspired Fluid-Structure Interaction, *"Reduced-Order Model of Hydrodynamically Interacting Bio-Inspired Wings".* Received travel support award (SIAM).
- 2023 Frontiers in Applied & Computational Mathematics (FACM 2023), NJIT; Conference on New Trends in Computational Wave Propagation and Imaging. *"Computationally Efficient Reduced-Order Models of Swimmers".*
- 2023 SIAM New York-New Jersey-Pennsylvania Section (NNP 2023), NJIT, NNP Annual Meeting;
- "*On Hydrodynamically Interacting Airfoils and Related Reduced-Order Models".*
- 2023 The Institute for Computational and Experimental Research in Mathematics (ICERM), Brown University, From Impact Factor to Influence Factor: Data Science and Policy for Social Justice Conference; "*Data Informed Machine Learning Enabled Genocide Detection and Data Storytelling."*

LEADERSHIP & PROFESSIONAL MEMBERSHIPS:

Professional Development: Princeton University Area Alumni Association VP Executive, YMCA Board of Directors Member, Princeton PACM Code@Nights and Hack Princeton experience, Center for Applied Mathematics and Statistics (CAMS) at NJIT webmaster, Society for Industrial and Applied Mathematics (SIAM), Society of Hispanic Professional Engineers (SHPE), Mathematics Association of America (MAA). **Campus/Community:** Additional tutoring experience as Math department Teaching Assistant including office hours; Kensington, Philadelphia Soup Kitchen volunteer, St. Paul's Catholic church.